

are formed and liberated than could possibly be accounted for by any dearth in the circulating blood caused by their departure from the blood-vessels, and it seems necessary to believe that some chemical substance circulating in the blood causes this great hyperplasia. If one reflects that a leucocytosis of 30,000 to 40,000 per cubic millimetre is not unusual, the colossal number of new leucocytes quickly formed and thrown into circulation becomes a matter of wonder.

The Wandering Cells.—It is clear that, in the adult animal, certain cells cling together and remain in a definite order to form a stable tissue which we recognize instantly and surely and expect to find always in its own appointed place. We are convinced of the specific character of the cells which make up such an organ as the liver and expect neither to find liver-cells in the kidney nor to find them giving rise to kidney-cells or thyroid-cells when they multiply. But the anatomist, knowing exactly what cells are proper to the structure of each organ, is never surprised to find other cells lying loose in the crevices of that tissue, and recognizes them as wandering cells not particularly at home anywhere, although they seem to have some favorite haunts. They are like the white cells of the blood and have the same powers of independent motility and phagocytosis. In the circulation they are hurried along, but in the interstices of the tissues they wander at leisure and pick up particles which attract them. There it is easy to recognize them by their sharp outline, the independence of attachment to other cells and, if they have not been allowed to grow chilled and retracted into a round form, by their pseudopods.

Of course their movements are relatively slow, but when they are photographed in a cinematographic film, at intervals of perhaps a minute, and then thrown on the screen at a rapid rate, as was done in the marvelous films of Dr. Carrell, Dr. Lewis and Dr. Canti, one acquires a completely new impression of cells. There the connective-tissue cells grow out majestically and smoothly from the margin of the field (which really represents a culture of tissues and cells *in vitro*), crossing and interlacing until a firm new structure is formed. Among these cells one may see others of quite different aspect worming their way with no thought of building. Arrived at the margin where they escape from the entanglement of these more serious fibroblasts, they show their true characters. Some are polymorphonuclear leucocytes and they hop about within a limited area in a sort of ecstatic frenzy, evidently throwing out and retracting pseudopods at a great rate. Then there are lymphocytes which move humbly, like slugs crawling only a little way with head to the ground. But also there are macrophages which reach out great arms, perhaps in two or more directions, and at the end of these arms there is a flourish of clear protoplasm with out-flung streamers that wave and search about for whatever can be seized or else the whole advancing margin of the cell flows out and comes back like a wave, sucking in any particle that comes in its way. If only we could see all cells in this way many difficulties would disappear. It is evident though, even without this revealing glimpse, that in the adult human being there are several kinds of mobile cells which

have special characters and functions which mark them out as perfectly distinct and recognizable individuals in the community. They are known by their size and the form of their nucleus, by their staining properties and the sort of granules they contain, by their movements, by the kind of things that stimulate and attract them, by their ability to absorb into their bodies granules or various chemical substances, and perhaps especially by the different ferments they produce.

But, although such individuals are recognized, no one seems content to say that each is merely the descendant of a long line of quite similar ancestors, as we would in the case of the cells of the liver or the in-



Fig. 70.—Early development of wandering cells in embryonic tissue (Maximow). Cross-section of a vessel of the area vasculosa with primitive blood-cells (*p. Blz*) in the lumen. Rounding off of endothelial cells (*m*) and their conversion into blood-cells; *Mz*, mesenchyme cells; *Ed*, endothelium.

testinal epithelium. Instead, there is an unceasing effort to trace their origin from some other sort of cell. This is partly because some of them, such as the red corpuscles and the neutrophile and eosinophile leucocytes, seem to be cells which are at the end of their course of development and specialization, and no longer able to divide and multiply into others like themselves, so that they must have ancestors of a different sort. Indeed, everyone agrees (and this is always a dangerous expression) that red corpuscles are directly derived from nucleated red cells, which do not come into the circulation, and that the leucocytes are directly derived from myelocytes. After that, however, all

is vague, and we do not venture to say that the other wandering cells cannot propagate their kind by division. The other difficulty is that all these cells are so ubiquitous and move about so much that it is hard to discover where they are produced. There is no such trouble about the cells of the liver or the spermatozoa—everyone knows exactly where to look for their new formation. It is true that it is pretty obvious that under ordinary conditions the bone-marrow is concerned in the production of red corpuscles and leucocytes, but if some tumor should spread so as to occupy most of the bone-marrow, blood formation would appear in the most unexpected places throughout the body. It is as though normally the formation of red corpuscles and leucocytes were finally assigned to the bone-marrow merely as a matter of economy in organization, for in the embryo it can be seen going on everywhere.

As to the mononuclear cells which have not so obviously run their course of development, there is greater difficulty. They are everywhere scattered loosely, or in small but recognizable accumulations, or even forming a regular tissue, but no one can say with certainty where lymphocytes are formed, nor where the larger mononuclear phagocytic cells especially arise. Every imaginable conception has been put forward and each has had its vogue. The distinctive characters of all these cells have become clearer in recent years, but the welter of theory about their origin and relations still prevails. Staining of dried smears or sections promised much, vital staining roused new hopes, culture of cells *in vitro* so that their activities and development may be traced, is in favor now, but still obscurities remain.

The cells concerned are about as follows: In the normal circulating blood beside the red corpuscles and the platelets (the latter supposed to be derivatives of the megaloeryocytes of the bone-marrow) there are several types of leucocytes. Of these the polymorphonuclear neutrophile leucocytes are most numerous and from 60 to 70 per cent of all the white cells. They are larger than red corpuscles, very active in their movements, with a nucleus usually in lobules united by narrow threads and cytoplasm which contains abundant minute neutrophile granules. Similar in form are the polymorphonuclear eosinophile leucocytes with large shining granules, which stain brightly with eosin. These are far less numerous and form not more than 1 per cent. Still less numerous are those of the same general conformation whose cytoplasm contains basophilic granules. Beside these there are lymphocytes which are sometimes smaller than red corpuscles, sluggish in motion with relatively large, round, deeply stained nucleus and no conspicuous granules in the scanty cytoplasm. Larger cells of similar character are called large lymphocytes. Another type, with large, rounded or indented vesicular nucleus and cytoplasm in which granules hardly appear by ordinary stains, but in which mitochondria and vitally staining granules can be found, are monocytes. These are actively motile phagocytes when they emigrate into the tissues. All of these cells can pass through the walls of the vessels and appear in the interstices of the tissues.

Outside the circulation in the tissues one finds ordinarily few stranded red corpuscles, except as the result of hæmorrhage or in inflamed tissues. Once out of the stream they end in disintegration. The fate of leucocytes which emigrate from the vessels, is less certain, for they may wander back again into the stream. But the polymorphonuclear neutrophiles and eosinophiles are not ordinarily found in the meshes of the normal tissues.

It is far otherwise with the mononuclear cells which are so abundant there. Lymphoid cells generally regarded as identical with the lymphocytes of the circulation are found wandering in many places. They are also generally regarded as identical with the cells which make up the denser tissue of the lymph-nodes, the Peyer's patches, Malpighian bodies of the spleen, etc., and are supposed to be formed in these situations. Larger pale cells are clustered in the middle of each of such lymphoid nodules, and it was Flemming's idea that these large cells produced the lymphoid cells. It is very uncertain, however, and the large cells look rather more like phagocytes and become conspicuously increased in the course of infections, but never seem to produce lymphocytes.

It is most difficult to determine the standing of these large mononuclear phagocytic cells which are distributed in the tissues, partly in some recognized relation but largely as free wandering cells. Even yet after a great deal of study with the most ingenious methods, there is no agreement as to their origin or their interrelations. They have been given many different names, such as *histiocytes*, *clasmatocyte*, *macrophage*, *reticulo-endothelial cell*, *adventitial cell*, *rhagiocrine cell*, *endothelial leucocytes*, *polyblast*, etc. We have ourselves hitherto called them mononuclear wandering cells and since this is quite non-committal we may go on with it but the terms *histiocyte* and *macrophage* are also used.

There are several groups of authors who hold different views as to the origin and relations of these cells which views may be followed in the huge literature on the subject, but no one seems to regard them as an independent type of cell constantly reproduced and originating from an ancestral cell of the mesenchyme, distinct from those which give rise to the other well recognized cells of the bone-marrow and blood, although this might seem the most reasonable idea. Briefly, Mallory and others derive them from endothelial cells. Naegeli, Piney and others relate them to myeloblasts. Maximow is sure that they develop from lymphocytes, Aschoff and Kiyono from reticulum cells and endothelium which they combine into the reticulo-endothelial system; Sabin, Doan and Cunningham by supra-vital stains separate sharply monocytes which occur in the tissues and also in the blood from clasmatocytes because monocytes have a circle of granules in the cytoplasm near the nucleus, which stain red with neutral red, while clasmatocytes have no such specific character. Naegeli finds that the monocytes give the oxidase reaction and have fine neutrophile granules which brings them close to the myeloid type rather than to the lymphocytes which have no oxidase reaction. Bloom, following Maximow, insists

that monocytes and lymphocytes both show the circle of red stained particles and ignores the difference in oxidase reaction. Lewis sees no real distinction between clasmatocytes and monocytes which Sabin and coworkers derive respectively from endothelium and reticular cells, and prefers to bring them together as one type. It seems that he really tends to go back to the very earliest stages in the mesenchyme for the origin of such mononuclear wandering cells, or macrophages.

It is well known that they abound in the tissues and are actively mobile so that they hasten to a place where their services are needed. They are abundant in the spleen, in lymph nodes, in the walls of the digestive tract and in the omentum and about blood vessels everywhere

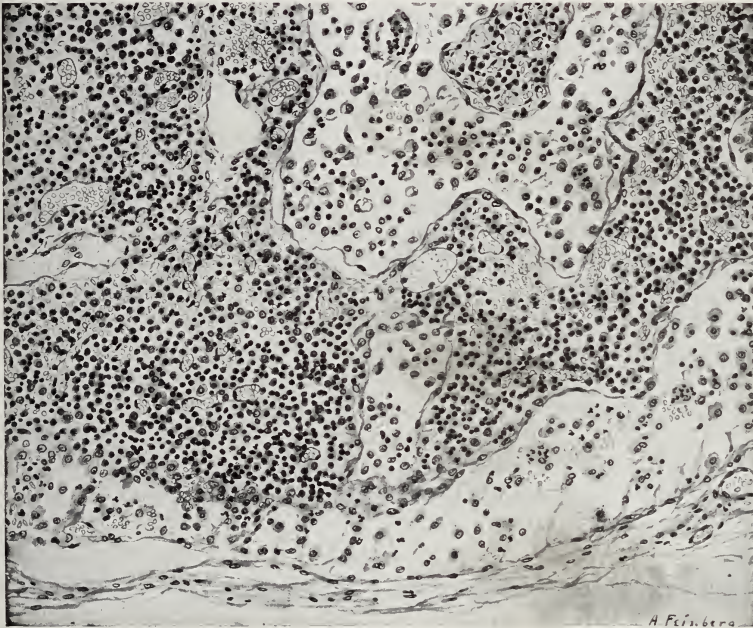


Fig. 71.—Lymph-gland showing phagocytic wandering cells in the lymph-sinuses.

and bronchi, and even in the septal tissues in the lung. It seems that cells identified with these by practically everyone are suspended or anchored within the endothelial lining of the blood sinuses of the liver. These are the Kupffer cells. Probably similar cells with similar anchorage are those which cross the peripheral sinuses in the lymph nodes and there are many quite analogous cells in the meshes of the tissue between the sinuses in the splenic pulp. Whether the actual lining cells of these blood sinuses of the spleen are, in part, of this character is questionable but it seems quite probable. At any rate, the outstanding character of this group of cells which we prefer to think of as a single type and derived from some remote ancestor in the mesenchyme, is their mobility with their phagocytic activity and apparently their important

rôle in relation with immunity and the protection of the body. In this connection the student should read the Harvey Lecture of Gay.

From the morphological point of view it is well to point out the distinctive features of such cells as they occur in different places. The monocyte in the circulating blood is a cell much larger than a lymphocyte with cytoplasm slightly basophilic, a nucleus usually bent on itself, not very deeply stained with very faint cytoplasmic granulations, sometimes showing the oxidase reaction and often enclosing fragments of red corpuscles or other outside material. Outside the circulation the same cell or its close relative is generally described as rather larger, with more active phagocytic nature, with perhaps paler cytoplasm and with no specific arrangement of neutral red staining granules. The Kupffer cells have long processes which suspend them along the walls of the liver sinuses. In the spleen one can in sections distinguish in the pulp large pale cells of irregular outline, loaded with phagocytized fragments, from other cells with more deeply stained nucleus and cytoplasm. It appears that these latter cells are derivatives of the lymphocytes from the Malpighian bodies, as Rich has recently demonstrated in artificial immune reactions.

Gay demonstrated the activity of the macrophages by irritating the pleura with an injection of aleuronat which brought them in great numbers to the subpleural tissues on that side after which he injected bacteria into the opposite pleura which brought them at once across to that side where they were actively protective.

These are, according to Gay, the cells which really confer protection by their presence rather than the polymorphonuclear neutrophiles which hurry so quickly to the point of infection.

The macrophages were studied by Metchnikoff who realized that they are little apt to phagocytise such bacteria as Streptococci and Pneumococci but do take up tubercle bacilli and some protozoa or fungi. They act vigorously as phagocytes for foreign substances and indeed one method of recognizing them is to inject into the blood stream carmine, India ink or some dye which then marks out brilliantly Kupffer cells and mononuclear phagocytes wherever they are normally lodged. It is interesting, as shown by Brickner and others, that in such animals the colored material is concentrated in the organs and tissues mentioned as the chief habitat of these cells while brain, heart muscles, etc., remain perfectly uncolored. The cells engulf foreign particles or partially surround them, if they are too large. Indeed, the cell itself increases in size to a great degree under such stimulus and with multiplication of nuclei becomes a giant cell, often then completely enclosing the foreign particle. It will be pointed out later that the tendency to engulf tubercle bacilli observed by Metchnikoff has led to the recognition of the fact that the epithelioid cells and giant cells of tubercles are derived from the monocytes or macrophages in the tissue. All of this can be made much clearer by reference to illustrations, if good ones can be made. The student is especially referred to the illustrations in Maximow's papers and in the histology of Maximow

and Bloom; also to the plates in the paper of Clough on "Monocytic Leukaemia," and the papers of Bloom and of Lang.

Lewis and his co-workers by the use of cultures of cells, seem however to have shown clearly, just as Maximow believed, that all the forms and sizes depend upon the age of the cell and its opportunity to stuff itself with foreign material and grow. Every stage up to the formation of huge giant-cells can be observed in these cultures, and the growth of a small mononuclear cell into a giant protoplasmic mass actually followed.

It is our impression that the endothelial cells which line the blood- and lymph-channels are limited to the fulfilment of that function. We have no evidence that they ever act as phagocytes, although foreign materials or phagocytic cells may adhere to them. Nor have we any convincing evidence that, once differentiated to the proper function, they ever divide the form wandering cells of any sort. Whether the formation of red corpuscles and leucocytes in the bone-marrow is ultimately dependent upon endothelial cells is open to grave doubt, in spite of the perfectly definite statements of so many writers. Always at the last stand they merely assume that endothelial cells must have been the source of the new cells. It seems more reasonable to suppose—although it is no great help—that at a very early stage in embryonic development ancestral cells were established for each of the types found in the adult, and that these, although inconspicuously lodged throughout the body, are still producing each its own type of cell. It does not seem reasonable to suppose that such cells would first undergo differentiation into endothelial cells or even angioblasts, which must then turn about and organize the ancestry for a whole series of different races of cells.

The explanations of the interrelations of these cells are so new and so little applied in human pathology that it may be excusable for a time to speak of all the large, phagocytic, mononuclear cells of the tissues simply as mononuclear wandering cells.

Among them there occurs one peculiar form which is probably derived from the lymphocyte through a modification of its nucleus and protoplasm. This is the so-called *plasma cell* described by Unna, which occurs normally in the intestinal mucosa and elsewhere and appears in great numbers in many forms of long-standing subacute inflammatory reactions (Fig. 74). Tuberculous granulation tissue and gonorrheal salpingitis afford examples of such conditions. The cells are rather larger than lymphocytes—somewhat amœboid, but in fixed preparations they usually assume a rounded or oval form, with the nucleus eccentrically placed, generally at one end of the cell. The nucleus, whose chromatin is in coarse masses, is surrounded by a pale halo, while the remaining protoplasm takes a bluish stain with ordinary nuclear dyes. Such cells, which may be regarded as a type slightly differentiated from the rest of the tribe of mononuclear wandering cells, are conspicuous because they are so constant in their form. They are not commonly phagocytic, but probably active in producing a digestive ferment.

The mere fact that all these cells take up in their nuclei or proto-

plasm or in their granules certain stains in a specific way does not, after all, afford us much information as to their true character. It would be more important to determine something as to their function.

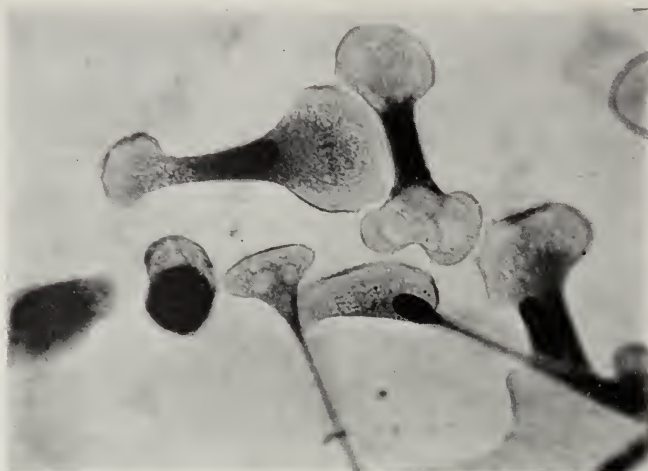


Fig. 72.—Monocytes from culture *in vitro* photographed while alive.

Metchnikoff, in his studies of inflammation and immunity, has been at great pains to show the analogy between the activities of the wandering cells of the body and those of amœbæ or other amœboid, simple,

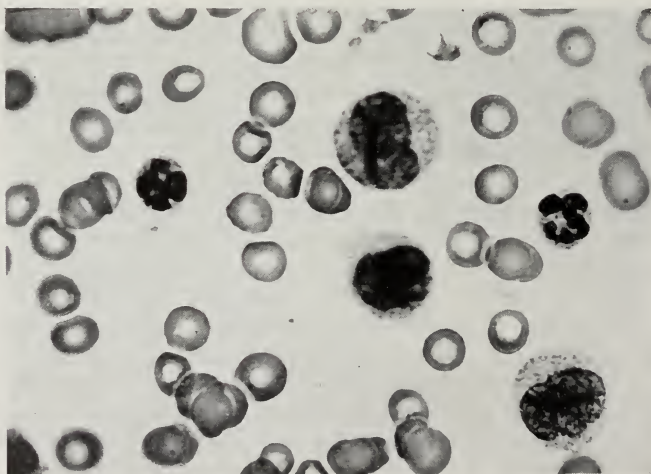


Fig. 73.—Monocytes in blood smear contrasting with polymorphonuclear leucocytes.

protoplasmic organisms. These latter swallow up bacteria and other substances, and by the aid of ferments or cytases digest them. These ferments, he says, are also present in the wandering cells of the body, and differ according to the type of cell, the small neutrophile leucocyte

possessing a so-called microcytase, while the mononuclear cells or macrophages, which eschew bacteria, except such as the tubercle and

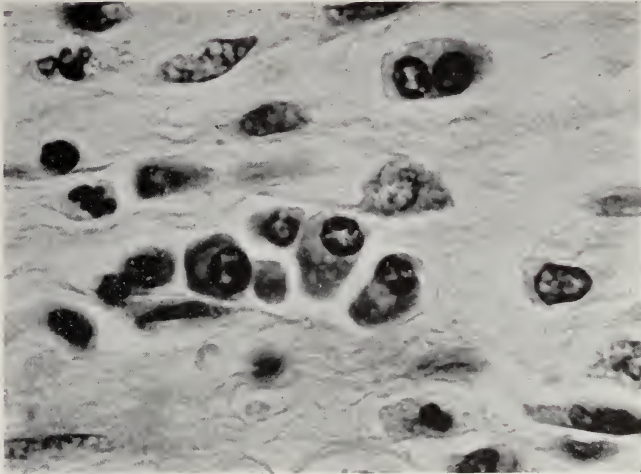


Fig. 74.—Cell infiltration in tissue lining a sinus leading from an old osteomyelitic abscess. In the centre are several plasma cells. About them are fibroblasts and three or four polymorphonuclear leucocytes.

leprosy bacilli, and digest with avidity cell débris, carry out their digestive processes by the aid of another ferment which he calls macrocytase.

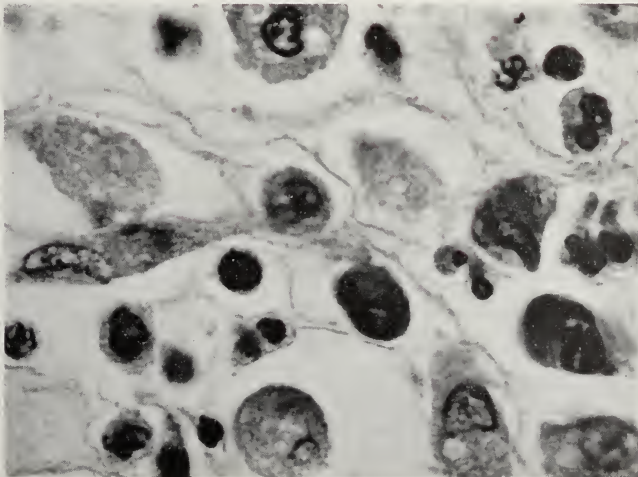


Fig. 75.—From the same tissue as Fig. 74. Macrophages, lymphoid cells, and polymorphonuclear leucocytes with fibroblasts and two plasma cells.

Many writers have recognized the power of certain tissues to digest themselves *in vitro* (autolysis), and have observed that the neutrophile

leucocytes produce a strong proteolytic ferment capable of digesting fibrin, gelatin, etc. Most of them, however, according to Wiens, have denied the production of a ferment by lymphoid cells. Opie has cleared the matter up very well by finding that the ferments of different cells require for their activity different reactions. He states that the polymorphonuclear neutrophile leucocytes and their ancestral granulated cells produce a trypsin-like ferment which acts best in an alkaline or neutral medium to digest proteins. Its action is often combated by an anti-enzyme, which is present in the plasma of the blood and in other body fluids. It is resistant to heat up to 70° or 75° C., and is therefore quite different from the complement of the serum, which is destroyed at 55° C. When formed in great concentration in a focal area of inflammation, the enzyme far outstrips the neutralizing anti-enzyme, and brings about the liquefaction of dead tissue, as in the case of an abscess. When in the presence of a great exudation of fluid, as in the pleural cavity, its action may be held in check. This proteolytic ferment Opie calls leucoprotease. Contrary to the results of other workers, he finds that, if the correctly feeble acid reaction be offered, the mononuclear cells also show the formation of a proteolytic ferment which is more like pepsin in its character, although it is unable to act in so strong an acid as is favorable to pepsin. This ferment, which he calls lymphoprotease, is produced in the lymph-nodes and in all accumulations of lymphoid cells, and of the various types of larger wandering mononuclear phagocytic cells which go to form Metchnikoff's group of macrophages.

While these ferments are evidently used inside the cell in the case of the particles which have been ingested, it seems certain that, in the formation of pus, as in the abscess, they are diffused from the bodies of the disintegrating leucocytes, and in the free fluid effect the solution of the adjacent injured and dead tissue. This function of the wandering cells is, of course, of immediate importance in connection with their task of cleaning up the injured area to prepare it for repair. While the proteases thus produced are active in the solution of undesirable material, their unbridled action might be detrimental. As a matter of fact, it is shown by Jobling and Petersen that the anti-ferment known to be present in the serum and to restrict the action of the ferment is a recognizable chemical substance, usually a soap or other combination of an unsaturated fatty acid. It is possible to remove or decompose this substance or to saturate the fatty acid with iodine and thus release the ferment to its full activity. The presence of excess of such soaps in the tubercle bacilli seems to be the cause of the delay of liquefaction of tissue brought to necrosis by those bacilli. Opie, however, points out that these experiments refer to leucoprotease, which is probably not concerned in the absorption of a tubercle or gumma. It is seen from this that we are at the beginning of our knowledge of the activities of the wandering cells. What other ferments they produce has been as yet only imperfectly studied, although we have evidence that others, such as oxydases, are produced by some of them, and there are surely more.